Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building reliable applications and automated test systems. Skilled in **C#, .NET, WPF**, Python, equipment integration, debugging, documentation, and quality-focused delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Software Engineering, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Structures, Computer Networks, Machine Learning

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Restructured fault-tolerant asynchronous workflows with explicit states, failure isolation, and background workers, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **software reliability** across 5K+ monthly production tasks.
- Diagnosed application and database bottlenecks for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through **root-cause analysis**, indexed queries, validation checks, and documented release procedures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Developed concurrent services for secure file ingestion and multimodal processing, handling 1K+ files per batch with fault isolation, provider fallbacks, and **software verification** across production grading workflows.
- Built operator-facing React and Flask interfaces supporting 150+ instructors and 10K+ daily requests, using REST APIs, automated checks, and **CI/CD deployment validation** to reduce operational turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*
- Built reproducible Python test pipelines processing 100K+ autonomous-driving logs, improving trajectory planning 18% while conducting **laboratory-style validation** through controlled simulations, baseline comparisons, failure analysis, and documented results.
- Evaluated a spatial-temporal decision system through closed-loop tests, improving stability 15% while investigating defects, measuring performance tradeoffs, and communicating **verification evidence** through technical reports and research reviews.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Engineered cross-platform software using object-oriented Singleton and Factory patterns, improving response times 30%, reducing technical debt 20%, and strengthening **maintainability** across distributed Windows, Linux, and macOS deployments.
- Automated 20+ **unit and integration tests** with PyTest, Mockito, Jenkins, and virtual-machine builds across three operating systems, supporting defect reproduction, corrective actions, Agile reviews, and release validation.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Built eight **C# and .NET** modules using object-oriented design for authentication, resources, billing, permissions, and notifications, reliably synchronizing 2M+ records across distributed locations with real-time operator updates.
- Implemented automated unit, integration, and end-to-end tests through Jenkins CI/CD and Docker staging, achieving 95%+ **build validation** coverage and reducing deployment time 50% across controlled production releases.

Skills

Languages: C#, C++, Python, Java, SQL, TypeScript, JavaScript, Bash
Desktop & Controls: .NET, WPF, MVVM, Windows Desktop, HMI, Control Logic, PLC Fundamentals, Industrial Automation, Equipment Integration, Hardware Interfaces, TCP/IP, UDP, Async/Await, OOP, Design Patterns
Testing & Quality: Unit Testing, Integration Testing, System Testing, Verification and Validation, Manual Testing, Test Plans, xUnit, PyTest, Regression Testing, Debugging, Root-Cause Analysis, Corrective Actions
Tools & Delivery: Git, CMake, Jira, Jenkins, GitHub Actions, CI/CD, Docker, REST APIs, PostgreSQL, SQL Server, SQLite, Release Notes, Requirements, Technical Documentation, Agile/Scrum, Code Reviews

Projects

Instrument Control Workbench | *C#, .NET 8, WPF, MVVM, Async/Await, TCP/IP, UDP, xUnit*
- Designed a Windows **equipment-control HMI** with asynchronous command queues, configurable device adapters, live telemetry, alarm handling, and xUnit-tested MVVM components for repeatable laboratory operation and commissioning workflows.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*
- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **control algorithms** with interactive visualization, validating 1K+ simulated process combinations and reducing average turnaround time up to 25%.

Chat2SQL | *Java, TypeScript, Docker, Maven, PostgreSQL, SQL Server*
- Extended a cross-platform workspace supporting 10+ databases, using Docker and Maven to validate **interface compatibility**, schema synchronization, and reproducible deployments across Windows, macOS, and Linux development environments.